**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 102 · Lesson 102 · Choosing RISC, CISC, GPU and Multicore Systems

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What does RISC primarily describe?

Q2. What does a GPU excel at?

Q3. Can a RISC CPU be multicore?

Q4. Graph edges A–B, A–C, B–D: BFS from A with B before C?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

ISA — Instruction Set Architecture.

accelerator — Hardware specialised for particular workloads.

latency — Delay of a task.

throughput — Amount of work per unit time.

## New learning

Understand — RISC/CISC and CPU/GPU are not a single four-way classification: a processor's ISA, core organisation and accelerators are different design properties.

Apply — A graphics or machine-learning workload with abundant parallel operations may benefit from a GPU; sequential/control-intensive jobs suit general CPU cores.

Evaluate — Choose based on power, energy budget, latency, software compatibility, transfer overhead and ability to divide the workload; support claims with mechanism.

## Worked example

Recommend hardware for battery-powered sensor making a small control decision each second.

1. Explain the choice — A low-power embedded CPU (often using a RISC architecture) is reasonable; no large GPU is needed for tiny sequential work.

# We Do · Practise with guidance

A studio renders millions of independent image elements per frame. Recommend computing hardware.

Explain the choice

A GPU can accelerate similar independent pixel/shading operations, paired with a CPU for control.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Explain why RISC/CISC and multicore are not competing mutually exclusive categories. [6]

### Question 2

Select suitable processor features for (a) small battery sensor (b) large parallel graphics processing. Justify. [6]

### Question 3

Evaluate CPU vs GPU investment for a school's interactive office tasks and high-volume video editing. [8]

# Exit Ticket · Answer before checking

Every multicore processor must use a CISC instruction set.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. A reduced/regular instruction-set design approach.

Q2. Large quantities of similar parallel operations.

Q3. Yes. ISA and core count are independent characteristics.

Q4. A,B,C,D.

### You Do pages · Independent questions

Q1. RISC/CISC describe ISA [2], multicore number of processing cores [2], a RISC or CISC implementation can be multicore [2].

Q2. Small low-power CPU/embedded processor [3]; GPU for many similar independent graphics operations [3].

Q3. Interactive CPU responsiveness [2], GPU parallel rendering benefit [2], power/cost/compatibility [2], balanced recommendation based on workload [2].

### Exit Ticket · Model answer

False. RISC processors can have multiple cores.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**